SIMATS ENGINEERING

Assessment - CO3-AT3

Subject: Artificial Intelligence

Course Code: CSA 1712

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Course Outcome Covered (CO3) — Resolution Algorithm Implementation

Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Question 1: Rain and Wet Ground

Knowledge Base: (1) If it rains, then the ground becomes wet. (2) It is raining.

Goal: Prove that the ground is wet.

1. Propositional Logic Representation

```
Let R = "It rains"
Let W = "The ground becomes wet"
Premise 1: R → W
Premise 2: R
```

2. Conversion to CNF

```
Clause 1 (from R → W):  ¬R ∨ W
Clause 2 (from R):        R
```

3. Negation of the Goal

```
Clause 3 (¬Goal, since Goal = W):  ¬W
```

4. Resolution Algorithm — Step by Step

Step	Clause 1	Clause 2	Resolve On	Resolvent
1	¬R ∨ W (Clause 1)	¬W (Clause 3)	W	¬R
2	¬R (Step 1 result)	R (Clause 2)	R	□ (Empty Clause)

5. Conclusion

Resolving Clause 1 with the negated goal (Clause 3) on W gives ¬R. Resolving this ¬R with Clause 2 (R) on R produces the empty clause □. Since resolution reaches □, the negation of the goal leads to a contradiction with the knowledge base, which proves the original goal: the ground is wet.

Question 2: Student Assignment Submission

Knowledge Base: (1) If a student submits the assignment, then the student receives internal marks. (2) Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks.

1. Propositional Logic Representation

```
Let S = "Rahul submits the assignment"
Let M = "Rahul receives internal marks"
Premise 1: S → M
Premise 2: S
```

2. Conversion to CNF

```
Clause 1 (from S → M):  ¬S ∨ M
Clause 2 (from S):        S
```

3. Negation of the Goal

```
Clause 3 (¬Goal, since Goal = M):  ¬M
```

4. Resolution Algorithm — Step by Step

Step	Clause 1	Clause 2	Resolve On	Resolvent
1	¬S ∨ M (Clause 1)	¬M (Clause 3)	M	¬S
2	¬S (Step 1 result)	S (Clause 2)	S	□ (Empty Clause)

5. Conclusion

Resolving Clause 1 with the negated goal on M gives ¬S. Resolving ¬S with Clause 2 (S) on S yields the empty clause □. The contradiction confirms the goal is proved: Rahul receives internal marks.

Question 3: Library Membership

Knowledge Base: (1) If a person is a library member, then the person can borrow books. (2) Priya is a library member.

Goal: Prove that Priya can borrow books.

1. Propositional Logic Representation

```
Let L = "Priya is a library member"
Let B = "Priya can borrow books"
Premise 1: L → B
Premise 2: L
```

2. Conversion to CNF

```
Clause 1 (from L → B):  ¬L ∨ B
Clause 2 (from L):         L
```

3. Negation of the Goal

```
Clause 3 (¬Goal, since Goal = B):  ¬B
```

4. Resolution Algorithm — Step by Step

Step	Clause 1	Clause 2	Resolve On	Resolvent
1	¬L ∨ B (Clause 1)	¬B (Clause 3)	B	¬L
2	¬L (Step 1 result)	L (Clause 2)	L	□ (Empty Clause)

5. Conclusion

Resolving Clause 1 with the negated goal on B gives ¬L. Resolving ¬L with Clause 2 (L) on L produces the empty clause □. No further clauses can be generated, and since □ has been derived, the goal is proved: Priya can borrow books.

Question 4: Placement Eligibility

Knowledge Base: (1) If a student clears the aptitude test, then the student is eligible for placement. (2) Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement.

1. Propositional Logic Representation

```
Let A = "Arun clears the aptitude test"
Let E = "Arun is eligible for placement"
Premise 1: A → E
Premise 2: A
```

2. Conversion to CNF

```
Clause 1 (from A → E):  ¬A ∨ E
Clause 2 (from A):        A
```

3. Negation of the Goal

```
Clause 3 (¬Goal, since Goal = E):  ¬E
```

4. Resolution Algorithm — Step by Step

Step	Clause 1	Clause 2	Resolve On	Resolvent
1	¬A ∨ E (Clause 1)	¬E (Clause 3)	E	¬A
2	¬A (Step 1 result)	A (Clause 2)	A	□ (Empty Clause)

Resolution Tree

```
      ¬A ∨ E         ¬E
            \       /    (resolve on E)
              ¬A          A
                 \       /    (resolve on A)
                    □
```

5. Conclusion

The resolution tree shows the negated goal (¬E) combining with Clause 1 to yield ¬A, which then combines with Clause 2 (A) to yield the empty clause □. Reaching □ proves, by contradiction, that the goal holds: Arun is eligible for placement.

Question 5: Access Control System

Knowledge Base: (1) If a user enters the correct password, then the user is authenticated. (2) If a user is authenticated, then the user is granted access. (3) The user entered the correct password.

Goal: Prove that the user is granted access.

1. Propositional Logic Representation

```
Let P = "The user enters the correct password"
Let A = "The user is authenticated"
Let G = "The user is granted access"
Premise 1: P → A
Premise 2: A → G
Premise 3: P
```

2. Conversion to CNF

```
Clause 1 (from P → A):  ¬P ∨ A
Clause 2 (from A → G):  ¬A ∨ G
Clause 3 (from P):        P
```

3. Negation of the Goal

```
Clause 4 (¬Goal, since Goal = G):  ¬G
```

4. Resolution Algorithm — Step by Step, With Explanation

Step	Clause 1	Clause 2	Resolve On	Resolvent
1	¬A ∨ G (Clause 2)	¬G (Clause 4)	G	¬A
2	¬P ∨ A (Clause 1)	¬A (Step 1 result)	A	¬P
3	¬P (Step 2 result)	P (Clause 3)	P	□ (Empty Clause)

Step 1: The negated goal ¬G is resolved with Clause 2 (¬A ∨ G) on the literal G, cancelling G and leaving ¬A — meaning "if the user is not authenticated, access is not granted" combines with "access is not granted" to conclude "the user is not authenticated."

Step 2: The result ¬A is resolved with Clause 1 (¬P ∨ A) on the literal A, cancelling A and leaving ¬P — i.e. "not authenticated" combined with "correct password implies authenticated" gives "the password was not entered correctly."

Step 3: The result ¬P is resolved with Clause 3 (P) on the literal P, cancelling both literals entirely and producing the empty clause □ — a direct contradiction, since Clause 3 states the password was entered correctly.

Resolution Tree

```
   ¬A ∨ G        ¬G
        \       /     (resolve on G)
   ¬P ∨ A     ¬A
```

```
         \       /      (resolve on A)
            ¬P         P
                 \   /    (resolve on P)
                   □
```

5. Final Conclusion

Since assuming the negation of the goal (¬G, "the user is not granted access") leads through valid resolution steps to the empty clause □, the assumption is contradictory given the knowledge base. Therefore the original goal is proved true: the user is granted access. This example also illustrates forward chaining in effect — password correctness chains through authentication to access — captured here formally through resolution refutation.